

Advanced approximate and symbolic computing

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Function interpolation and approximation V.

Uniform approximation

Uniform approximation

- TASK: Given $f(x) \in C[a, b]$ continuous function, give its polynomial approximation with $p(x) = \sum_{i=0}^n \alpha_i x^i$ which is taken from the linear space $\mathcal{P}_n = \text{span}(1, x, x^2, \dots, x^n)$.
- Now we consider the space $C[a, b]$ with

$$\|g(x)\|_{\infty} \stackrel{\text{def}}{=} \max_{x \in [a, b]} |g(x)|$$

maximum- or Chebyshev-norm as normed linear space.

- This is called *uniform- or minimax- or Chebyshev-approximation*.

Uniform approximation

- Given $f(x) \in C[a, b]$ continuous function, among its approximations of form $p(x) = \sum_{i=0}^n \alpha_i x^i$ find the *best approximation* polynomial $p^*(x)$ such that

$$\|f(x) - p^*(x)\|_{\infty} \leq \|f(x) - p(x)\|_{\infty} \quad (1)$$

holds for all polynomials $p(x) \in \mathcal{P}_n$.

- The existence of the polynomial $p^*(x) \in \mathcal{P}_n$ follows from our previous theorem.
- However, the Chebyshev-norm is not strictly convex, the unicity is not granted.

Example: Chebyshev-norm is non strictly convex

- Consider functions $f(x) \equiv 1$ and $g(x) = x$ as elements of the linear space $C[-1, 1]$.
- $\|f(x)\|_{\infty} = \|g(x)\|_{\infty} = 1$
- For strictly convex norms it holds that

$$\frac{1}{2}\|f(x) + g(x)\|_{\infty} < 1.$$

- However, we have

$$\frac{1}{2}\|f(x) + g(x)\|_{\infty} = \max_{x \in [-1, 1]} \frac{|x + 1|}{2} = 1.$$

Example: finding $p^*(x)$ by hand

- Find the uniform best approximating 0th degree polynomial of the function $f(x) = \exp(x)$ on the interval $[0, 1]$.
- Thus, $p^*(x)$ is a constant function: $p^*(x) = c_0$.
- Since $f(0) = 1$ and $f(1) = \exp(1) = e$, we have $1 \leq c_0 \leq e$.
- Moreover, we know that $f(x)$ is monotonically increasing, p^* is constant, thus $f - p^*$ is also monotonically increasing.
- Hence, $\max_{x \in [0, 1]} |f(x) - p^*(x)|$ is taken at an extreme point. Thus

$$\|f - p^*\|_\infty = \max\{|\exp(0) - c_0|, |\exp(1) - c_0|\}.$$

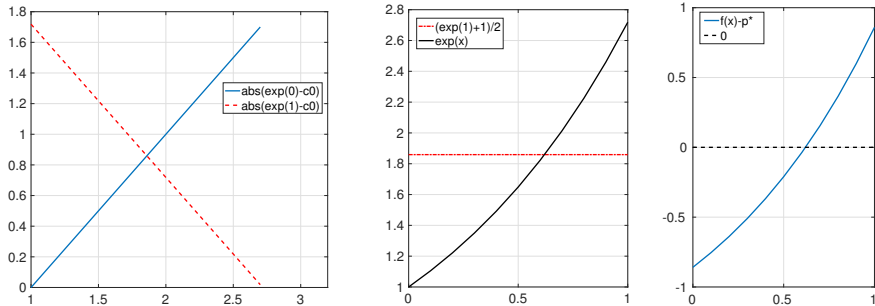


Figure 1: Left: the crossing point gives the maximum, thus $p^* = c_0 = (1 + \exp(1))/2$. Center: $\exp(x)$ and the obtained constant function: p^* is first *above* and then later it is *below* the function $f(x)$. Right: the difference of the two functions, we have different signs at the endpoints.

- We certainly have noticed the oscillating behavior in the example above.
- This is a general property for uniform approximation.
- The exact claim is given by the following theorem.

Theorem 12.1 (*de la Vallée-Poussin*) Let $f \in C[a, b]$ and assume that $r \in \mathcal{P}_n$ is such a polynomial for which there exist $n + 2$ points $\{x_j\}_{j=0}^{n+1}$ such that $a \leq x_0 < \dots < x_{n+1} \leq b$, where

$$\operatorname{sgn}(f(x_j) - r(x_j)) = -\operatorname{sgn}(f(x_{j+1}) - r(x_{j+1})) \quad (j = 0, \dots, n).$$

Then we have

$$\min_{p \in \mathcal{P}_n} \|f - p\|_\infty \geq \min_{0 \leq j \leq n+1} |f(x_j) - r(x_j)|.$$

the $\operatorname{sgn}(x)$ is the signum function: the value is $+1$ if $x > 0$ and -1 if $x < 0$.

Chebyshev oscillation theorem

- By the previous theorem we give a *lower bound* for the error of the uniform approximation.
- Do we have such polynomial? Chebyshev said yes.

Theorem 12.2 *Let $f \in C[a, b]$. The polynomial $p^* \in \mathcal{P}_n$ is uniform best approximation if and only if there exist $n + 2$ points $x_0 < x_1 < \dots < x_{n+1}$ such that*

$$|f(x_j) - p^*(x_j)| = \|f - p\|_\infty \quad (j = 0, \dots, n + 1)$$

and the sign of the error oscillates in these point, i.e.,

$$f(x_j) - p^*(x_j) = -(f(x_{j+1}) - p^*(x_{j+1})) \quad (j = 0, \dots, n).$$

- It is important to mention that the polynomial $p^* \in \mathcal{P}_n$ uniquely exists.

Application: optimal interpolation with Chebyshev polynomials

- Interpolation???
- We are looking for the base points for which

$$\|f - p_n\|_\infty$$

is minimal, where $p_n \in \mathcal{P}_n$ polynomial interpolates the function f in the given bases points.

- Interpolation: find the polynomial given the basis points.
- Here: find the optimal basis points.

Application: optimal interpolation with Chebyshev polynomials

- A fact from Lagrange interpolation theory: If f is $n + 1$ -times continuously differentiable on the interval $[a, b]$, then for all $x \in [a, b]$ there exists such $\xi \in [a, b]$ for which

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)(x - x_1) \dots (x - x_n).$$

- Take the absolute value of both sides and find the maximum on the interval $[a, b]$:

$$\|f - p_n\|_\infty = \max \frac{|f^{(n+1)}(\xi)|}{(n+1)!} \max_{x \in [a, b]} |(x - x_0)(x - x_1) \dots (x - x_n)|.$$

- We need to concentrate on the 2nd term only.

- More precisely, we would like to find the minimum of it:

$$\min_{x_0, \dots, x_n} \max_{x \in [a, b]} |(x - x_0)(x - x_1) \dots (x - x_n)| \quad (2)$$

- Note that

$$(x - x_0)(x - x_1) \dots (x - x_n) = x^{n+1} - r_n(x)$$

for some polynomial $r_n \in \mathcal{P}_n$ which coefficients depend on the basis points $\{x_j\}_{j=0}^n$.

- For example $n = 1$:

$$(x - x_0)(x - x_1) = x^2 - ((x_0 + x_1)x - x_0x_1) = x^2 - r_1(x),$$

here by changing x_0 and x_1 we obtain different polynomial $r_1(x)$.

Application: optimal interpolation with Chebyshev polynomials

- In order to obtain the optimal interpolation points we need to solve

$$\min_{r \in \mathcal{P}_n} \max_{x \in [a,b]} |x^{n+1} - r(x)| = \min_{r \in \mathcal{P}_n} \|x^{n+1} - r(x)\|_{\infty}.$$

- Thus we need to approximate a polynomial of degree $n + 1$ with a polynomial of degree n .

Chebyshev polynomial

- Let $[a, b] = [-1, 1]$.
- The definition of *Chebyshev polynomial*:

$$T_n(x) = \cos(n \cos^{-1}(x)),$$

which can be written as a recurrent formula:

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x),$$

where $T_0(x) = 1$ and $T_1(x) = x$.

- The Chebyshev polynomials form an orthogonal polynomial system with the weight function $\rho(x) = 1/\sqrt{1-x^2}$.

Chebyshev polynomial

- Despite we have trigonometric functions in the definition these are polynomials:

$$n = 0 : \quad T_0(x) = \cos(0 \cos^{-1} x) = \cos(0) = 1$$

$$n = 1 : \quad T_1(x) = \cos(\cos^{-1} x) = x$$

$$n = 2 : \quad T_2(x) = \cos(2 \cos^{-1} x) = 2 \cos^2(\cos^{-1} x) - 1 = 2x^2 - 1$$

$$\vdots$$

Chebyshev polynomial

- The function $T_n(x)$ oscillates for different n : this is one of the properties we can exploit.
- Since the leading coefficient of $T_{n+1}(x)$ is 2^n if $n \geq 0$, thus we introduce the *normalised Chebyshev polynomials*:

$$\hat{T}_{n+1}(x) = 2^{-n}T_{n+1}(x) \quad (n \geq 0),$$

where $\hat{T}_0(x) = 1$.

- Due to the normalization our polynomials are of form

$$\hat{T}_{n+1}(x) = x^{n+1} - r(x)$$

for some polynomial $r(x) \in \mathcal{P}_n$.

- In fact the polynomial $r(x)$ is not that interesting. Instead the root of $\hat{T}_{n+1}$ can be calculated as

$$\eta_j = \cos \left(\frac{j\pi}{n} \right) \quad (j = 0, \dots, n).$$

- These are the **optimal interpolation basis points**.

- Practice tells us that if the derivative of f is bounded (or it has some smoothness property), then the interpolation made in the Chebyshev basis points has generally very good result.

Chebyshev polynomials

- So far we have had the interval $[-1, 1]$, but of course the results can be generalized for arbitrary interval $[a, b]$: use the transformation for the basis points x_k :

$$\hat{x}_k = \frac{a+b}{2} + \frac{b-a}{2}x_k.$$

EXAMPLE: Runge function

- We revisit the famous Runge function:

$$f(x) = 1/(x^2 + 1)$$

on the interval $[a, b] = [-5, 5]$.

- Applying Lagrange interpolation on equidistant basis points results in a highly oscillating polynomial towards the end points.
- However, if we choose the basis points to be the optimal Chebyshev ones, the result is way much better.

- Let us see two different n values.

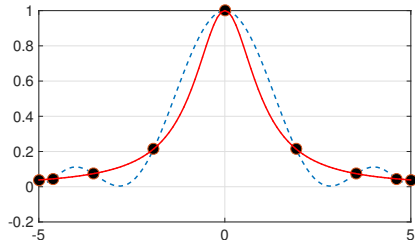
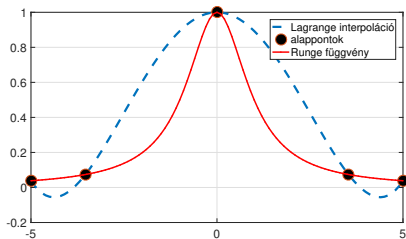


Figure 2: Left: $n = 4$, right: $n = 8$.

- Increasing n makes the interpolation even better.
- Note that the optimal basis points become denser (closer to each other) towards the end points.

